Assignment - 6

Interpolation & Extrapolation

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Problem Description

A pressure sensor is mounted on a piston-cylinder apparatus filled with a gas. The piston is moved quasi-statically so that the gas remains approximately isothermal. The sensor produces a voltage signal V that depends non-linearly on the true gas pressure P. No functional form relating voltage, pressure or volume is assumed unless explicitly stated.

PART 1

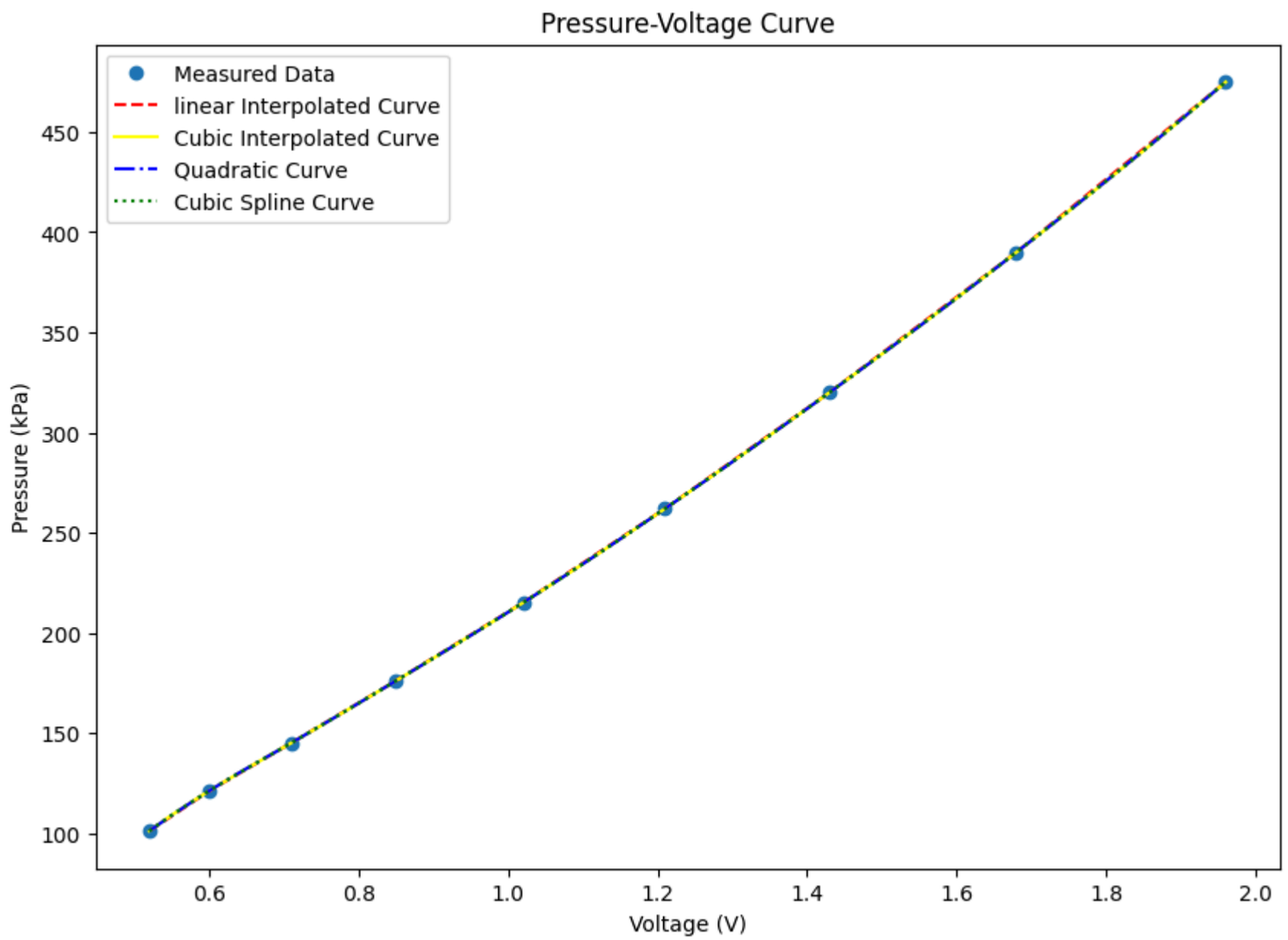
Voltage V (V)	Pressure P (kPa)
0.52	101
0.60	121
0.71	145
0.85	176
1.02	215
1.21	262
1.43	320
1.68	390
1.96	475

Table 1: Measured voltage and corresponding pressure values

In [54]:
```
#importing required libraries
import numpy as np
import matplotlib.pyplot as plt
from scipy.interpolate import interp1d, CubicSpline
```

In [55]:
```
#Measured voltage and corresponding pressure values
V = np.array([0.52, 0.60, 0.71, 0.85, 1.02, 1.21, 1.43, 1.68, 1.96])  # Voltage (V)
P = np.array([101, 121, 145, 176, 215, 262, 320, 390, 475])           # Pressure (kPa)
```

```python
v = np.linspace(min(V), max(V), 100)

f_linear = interp1d(V, P, kind='linear', fill_value="extrapolate")
f_cubic  = interp1d(V, P, kind='cubic',  fill_value="extrapolate")
f_quadratic = interp1d(V, P, kind='quadratic', fill_value="extrapolate")
cs_func = CubicSpline(V, P)

#functional representation of pressure P(V)
p_lin = f_linear(v)  #linear function
p_cube = f_cubic(v)  #cubic function
p_quad = f_quadratic(v)  #quadratic function
p_cs = cs_func(v)   #CubicSpline function

#plot graph
plt.figure(figsize=(10,7))
plt.plot(V, P, 'o', label='Measured Data')
plt.plot(v, p_lin, '--', label='linear Interpolated Curve', color='red')
plt.plot(v, p_cube, '-', label='Cubic Interpolated Curve', color='yellow')
plt.plot(v, p_quad, '-.', label='Quadratic Curve', color='blue')
plt.plot(v, p_cs, ':', label='Cubic Spline Curve', color='green')
plt.xlabel('Voltage (V)')
plt.ylabel('Pressure (kPa)')
plt.title('Pressure-Voltage Curve')
plt.legend(loc="best")
plt.show()
```



Observations

so, we can say from graph, measured data from sensor appear linear with respect to the pressure. here we used interp1d(linear, cubic, quadratic) and CubicSpline method to interpolate.

The data show a monotonic increasing trend.The relationship is approximately linear over the calibration range.However, slight curvature is visible so the response is not perfectly linear.

The dataset is sufficient to construct a calibration curve, but the functional form is not unique and depends on the interpolation method used.

A physically reasonable calibration curve should be smooth, continuous, monotonic, and free from oscillations, ensuring physically realistic behavior between measured points.

PART 2

Operating beyond the calibration

Voltage V (V)	Pressure P (kPa)
0.47	96
0.50	96
0.50	505
2.05	505
2.15	540
2.30	610

In [56]:
```
#Voltage signal "outside" the calibration data
V_out = np.array([0.47, 0.50, 2.05, 2.15, 2.30])
#reference pressure values
P_ref = np.array([96, 99, 505, 540, 610])

#P_out calculated from the CubicSpline extrapolation
P_out = cs_func(V_out)

#P_out calculated from the interp1d extrapolation
P_out_lin  = f_linear(V_out)
P_out_cube  = f_cubic(V_out)
P_out_quad = f_quadratic(V_out)

#P_out values calculated from the functional represented of pressure
print(f"P_ref: {P_ref}")
print(f"P_out: {P_out}")
print(f"P_out_lin: {P_out_lin}")
print(f"P_out_cube: {P_out_cube}")
print(f"P_Out_quad: {P_out_quad}")
```

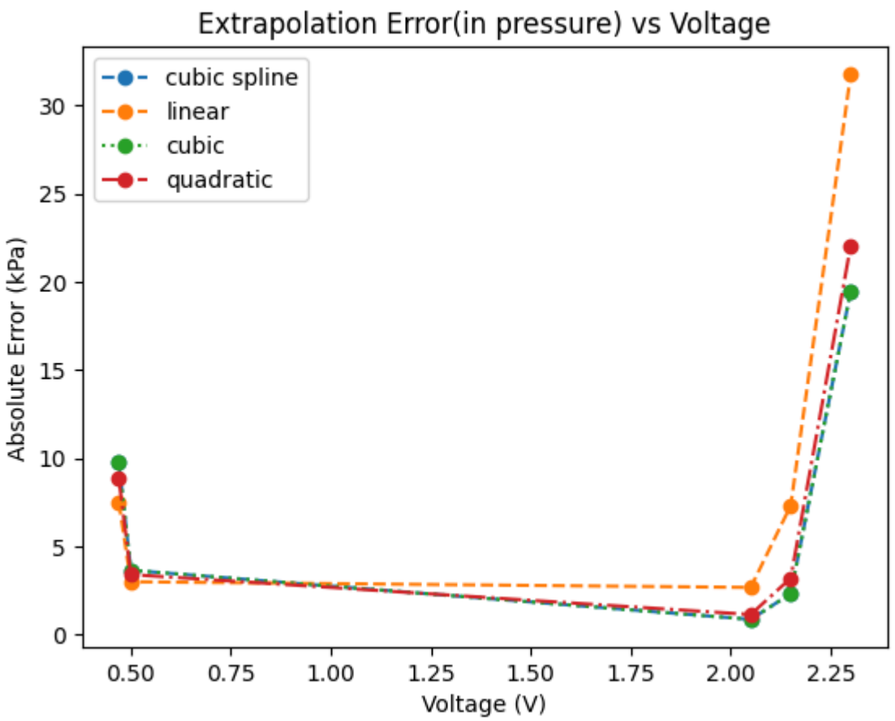
```
P_ref: [ 96  99 505 540 610]
P_out: [ 86.2333803   95.34789021 504.13146043 537.6748178  590.51226329]
P_out_lin: [ 88.5         96.         502.32142857 532.67857143 578.21428571]
P_out_cube: [ 86.2333803   95.34789021 504.13146043 537.6748178  590.51226329]
P_Out_quad: [ 87.17898959  95.59353526 503.85676902 536.79586578 587.93349792]
```

In [57]:
```
#error analysis
err_cs = np.abs(P_out - P_ref)
err_lin = np.abs(P_out_lin - P_ref)
err_cube = np.abs(P_out_cube - P_ref)
err_quad = np.abs(P_out_quad - P_ref)

#print error
print(f"error_cs: {err_cs}")
print(f"error_lin: {err_lin}")
print(f"error_cub: {err_cube}")
print(f"error_quad: {err_quad}")
```

```
error_cs: [ 9.7666197   3.65210979  0.86853957  2.3251822  19.48773671]
error_lin: [ 7.5         3.          2.67857143  7.32142857 31.78571429]
error_cub: [ 9.7666197   3.65210979  0.86853957  2.3251822  19.48773671]
error_quad: [ 8.82101041  3.40646474  1.14323098  3.20413422 22.06650208]
```

In [58]:
```
#Extrapolation Error Analysis
plt.plot(V_out, err_cs, '--o', label="cubic spline")
plt.plot(V_out, err_lin, '--o', label="linear")
plt.plot(V_out, err_cube, ':o', label="cubic")
plt.plot(V_out, err_quad, '-.o', label="quadratic")
```

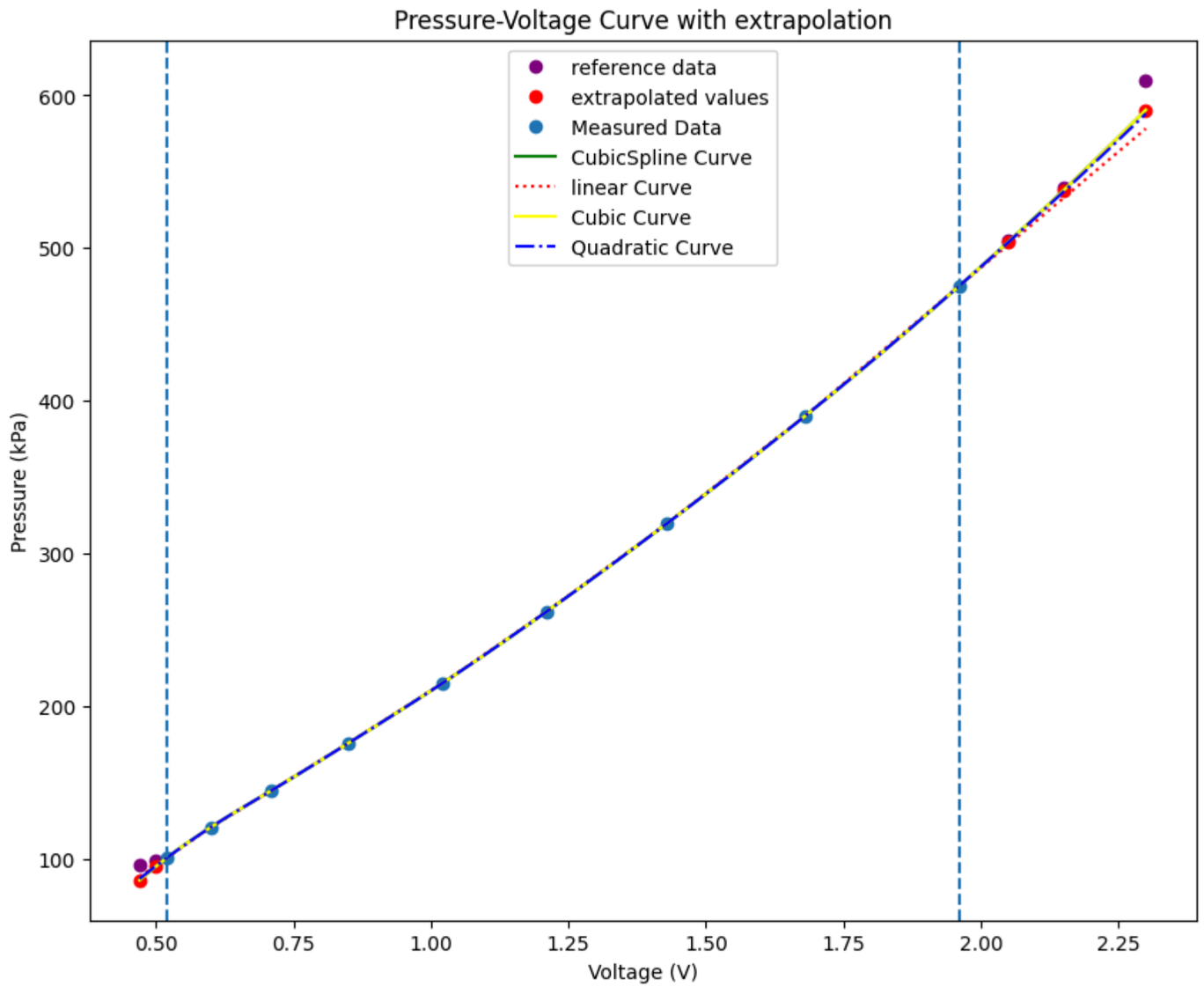
```
plt.xlabel("Voltage (V)")
plt.ylabel("Absolute Error (kPa)")
plt.title("Extrapolation Error(in pressure) vs Voltage")
plt.legend(loc="best")
plt.show()
```



Based on comparison with reference values, cubic spline extrapolation provides the most reliable estimates with minimal error near the calibration boundaries. Linear and quadratic methods exhibit larger deviations as the extrapolation distance increases. This highlights the inherent uncertainty associated with extrapolation beyond measured data.

In [59]:
```
v_ext = np.linspace(min(V_out), max(V_out), 100)
p_ext = cs_func(v_ext)
p_ext_lin = f_linear(v_ext)
p_ext_cube = f_cubic(v_ext)
p_ext_quad = f_quadratic(v_ext)


plt.figure(figsize=(10,8))

plt.plot(V_out, P_ref, 'o', color='purple', label='reference data') #reference data
plt.plot(V_out, P_out_cube, 'o', color='red', label='extrapolated values'
)  #extrapolated data

plt.plot(V, P, 'o', label='Measured Data')
plt.plot(v_ext, p_ext, '-', label='CubicSpline Curve', color='green')
plt.plot(v_ext, p_ext_lin, ':', label='linear Curve', color='red')
plt.plot(v_ext, p_ext_cube, '-', label='Cubic Curve', color='yellow')
plt.plot(v_ext, p_ext_quad, '-.', label='Quadratic Curve', color='blue')

#To demonstrate extrapolation clearly, drawing calibration range limits
plt.axvline(0.52, linestyle="--")
plt.axvline(1.96, linestyle="--")
```

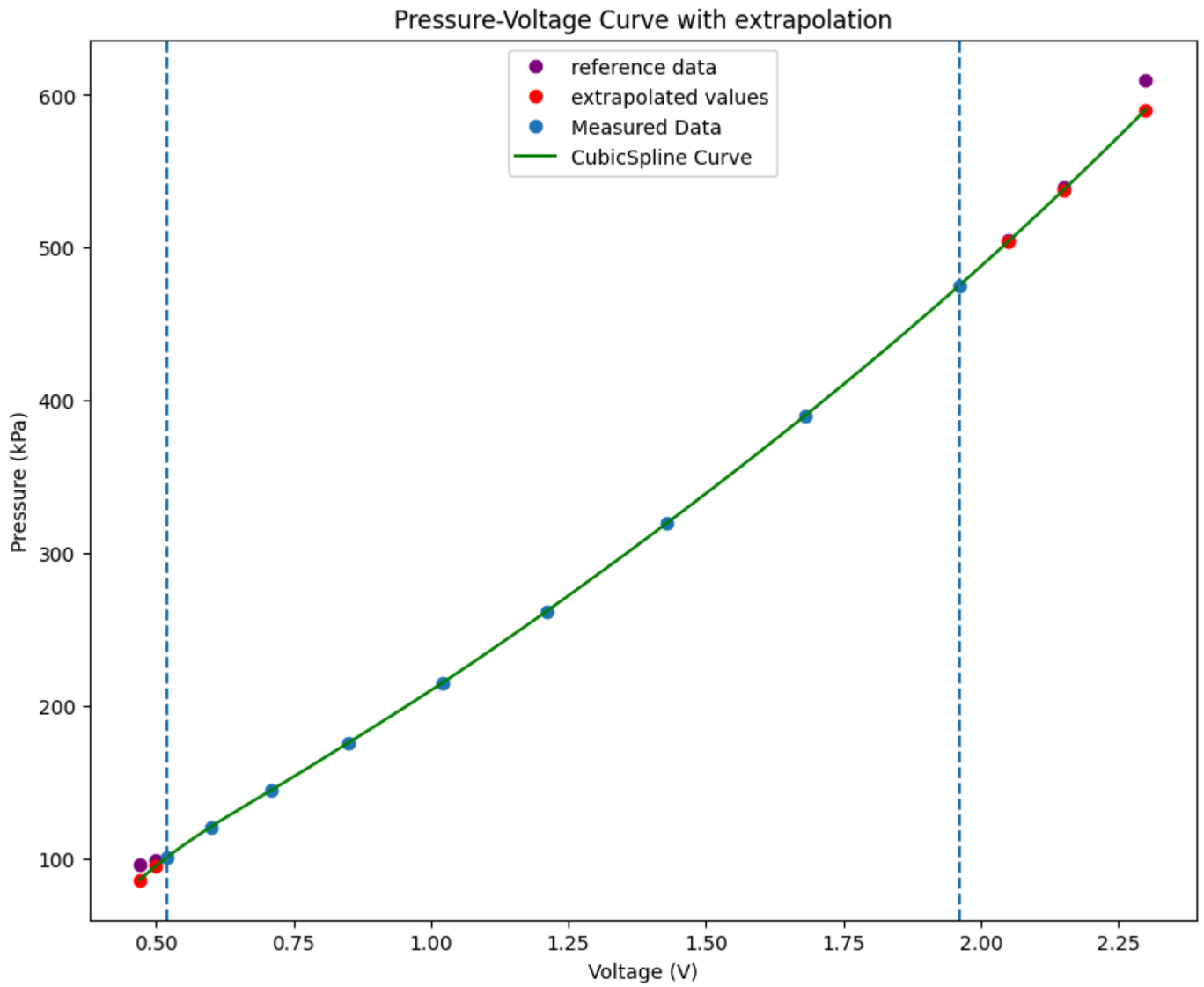
```
plt.xlabel('Voltage (V)')
plt.ylabel('Pressure (kPa)')
plt.title('Pressure-Voltage Curve with extrapolation')
plt.legend(loc='upper center')
plt.show()
```



In [60]:

```
#now plot only for cubicSpline extrapolated curve to observe clearly
plt.figure(figsize=(10,8))
plt.plot(V_out, P_ref, 'o', color='purple', label='reference data') #reference data
plt.plot(V_out, P_out_cube, 'o', color='red', label='extrapolated values'
)  #extrapolated data

plt.plot(V, P, 'o', label='Measured Data')
plt.plot(v_ext, p_ext, '-', label='CubicSpline Curve', color='green')

#To demonstrate extrapolation clearly, drawing calibration range limits
plt.axvline(0.52, linestyle="--")
```

```
plt.axvline(1.96, linestyle="--")

plt.xlabel('Voltage (V)')
plt.ylabel('Pressure (kPa)')
plt.title('Pressure-Voltage Curve with extrapolation')
plt.legend(loc='upper center')
plt.show()
```



> Smoother interpolation schemes (such as cubic splines) generally produce better extrapolation near the calibration range because they preserve slope continuity. However, accuracy deteriorates as the extrapolation distance increases.

> Extrapolation becomes more uncertain as we move farther away from the calibration range. Predictions near the known data points may still be reasonable, but far outside the measured region the error can grow quickly, making the results less reliable. Since extrapolation predicts behavior outside the measured range, the resulting curve may be numerically smooth yet physically inaccurate. A well-behaved mathematical curve does not guarantee physical validity.

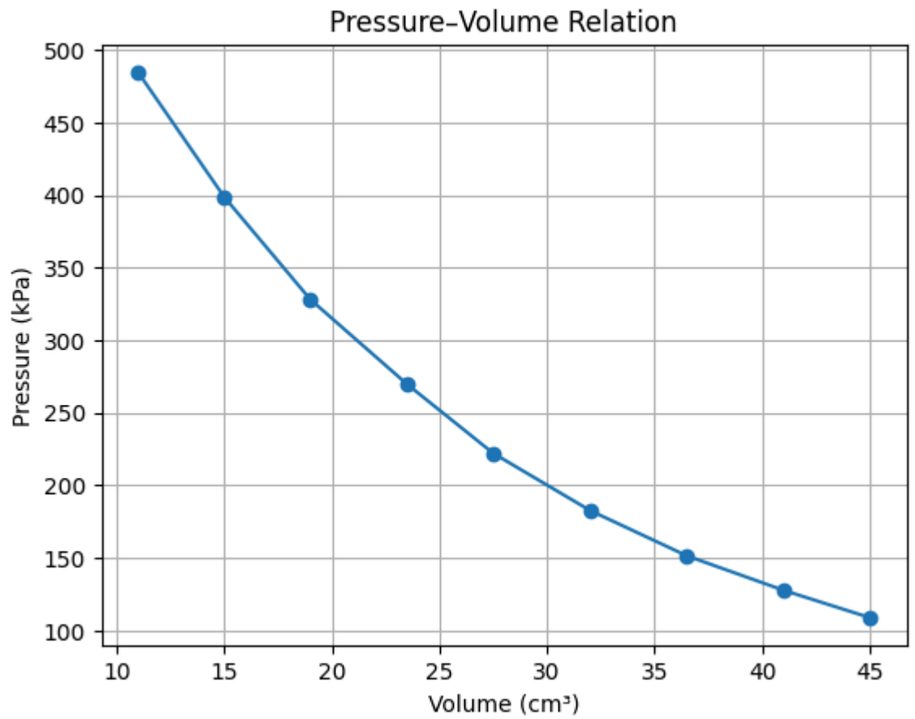
PART 3

Gas volume$V$ is related to the piston displacement by:

$$V(x) = V_0 - Ax$$

where $V_0 = 50cm^3$ and $A = 5cm^2$

In [61]:
```
#Measured voltage and corresponding displacement values
Volt_V = np.array([0.55, 0.63, 0.74, 0.88, 1.05, 1.24, 1.46, 1.71, 1.99]) #V
Disp_X = np.array([1.0, 1.8, 2.7, 3.6, 4.5, 5.3, 6.2, 7.0, 7.8])  #Cm


V_0 = 50  #initial volume (cm^3)
A = 5     #piston area (cm^2)

#Defining function for calculating Volume
def gas_volume(V_0, A, Disp_X):
  return V_0 - A * Disp_X

Volume = gas_volume(V_0, A, Disp_X)

#using CubicSpline function(calibration curve) for calculating the pressure from voltage measurement
Pressure = cs_func(Volt_V)

#other functions to calculate pressure
Pressure_linear = f_linear(Volt_V)
Pressure_cubic = f_cubic(Volt_V)
Pressure_quad = f_quadratic(Volt_V)

plt.plot(Volume, Pressure, 'o-')
plt.xlabel('Volume (cm³)')
plt.ylabel('Pressure (kPa)')
plt.title('Pressure-Volume Relation')
plt.grid(True)
plt.show()

#results
print(f"Volume: {Volume}")
print(f"Pressure: {Pressure}")
print(f"Pressure calculated using linear function: {Pressure_linear}")
print(f"Pressure calculated using Cubic function: {Pressure_cubic}")
print(f"Pressure calculated using quadratic function: {Pressure_quad}")
```




```
Volume: [45.  41.  36.5 32.  27.5 23.5 19.  15.  11. ]
Pressure: [108.93244278 127.74799715 151.51622358 182.75776669 222.16708847
 269.71161414 328.15906098 398.74011728 484.60375442]
Pressure calculated using linear function: [108.5        127.54545455 151.64285714 182.88235294 222.42105263
 269.90909091 328.4        399.10714286 484.10714286]
Pressure calculated using Cubic function: [108.93244278 127.74799715 151.51622358 182.75776669 222.16708847
 269.71161414 328.15906098 398.74011728 484.60375442]
Pressure calculated using quadratic function: [108.80484856 127.82933318 151.52592929 182.75901789 222.17393149
 269.70064842 328.16108631 398.76134546 484.53593163]
```

> For an ideal isothermal process, pressure and volume are inversely related. Using the calibration curve to convert voltage into pressure and the displacement data to compute volume, the reconstructed P–V curve is smooth and physically consistent. Although the curve does not show a perfect inverse relation, it reflects realistic experimental behavior.

> The obtained pressure–volume relation is physically reasonable and free from abrupt or unphysical variations. Deviations from the ideal isothermal relation may arise from experimental conditions, sensor calibration, and numerical interpolation.

PART 4

Suppose the gas approximately follows a polytropic relation:

$$PV^{\gamma} = constant$$

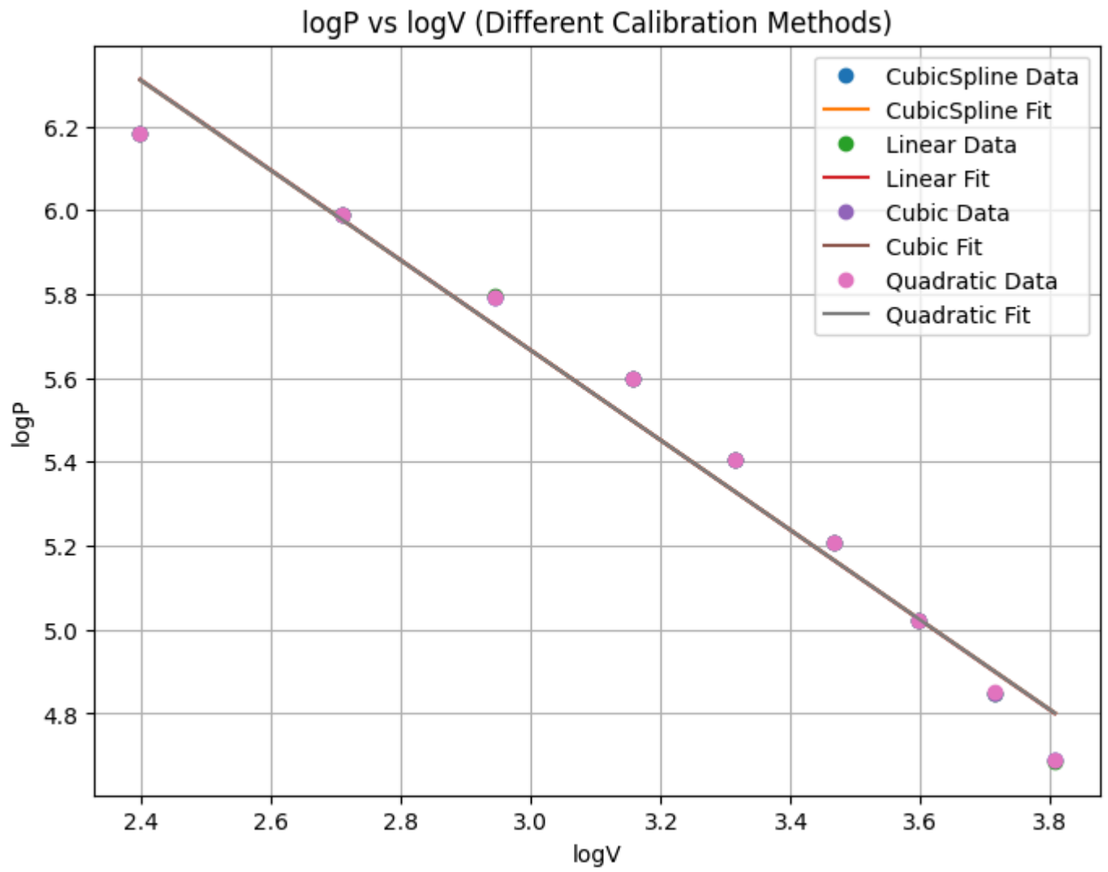
take log both side

$$log(P) + \gamma log(V) = log(C)$$

$$log(P) = -\gamma log(V) + log(C)$$

In [62]:

```
#Take log of volume
logV = np.log(Volume)

#Store all pressure datasets in a dictionary
pressure_models = {
    "CubicSpline": Pressure,
    "Linear": Pressure_linear,
    "Cubic": Pressure_cubic,
    "Quadratic": Pressure_quad
}

gamma_values = {}

plt.figure(figsize=(8,6))

for name, P_data in pressure_models.items():

    logP = np.log(P_data)

    #Linear fit in log-log space
    slope, intercept = np.polyfit(logV, logP, 1)

    gamma = -slope
    gamma_values[name] = gamma

    #Fitted straight line
    logP_fit = slope * logV + intercept

    #Plot for each function
    plt.plot(logV, logP, 'o', label=f'{name} Data')
    plt.plot(logV, logP_fit, '-', label=f'{name} Fit')

#plot
plt.title("logP vs logV (Different Calibration Methods)")
plt.xlabel("logV")
plt.ylabel("logP")
plt.legend()
plt.grid(True)
plt.show()

#print gamma values
print("\nExtracted gamma values:")
for name, gamma in gamma_values.items():
    print(f"{name}: γ = {gamma:.7f}")
print(f"\nSpread in γ = {max(gamma_values.values()) - min(gamma_values.values()):.5f}")
```




```
Extracted gamma values:
CubicSpline: γ = 1.0715270
Linear: γ = 1.0728388
Cubic: γ = 1.0715270
Quadratic: γ = 1.0716638

Spread in γ = 0.00131
```

> γ was obtained by fitting a straight line to the logP vs logV plot. Because pressure values depend on the interpolation method used for calibration, the inferred γ may vary slightly with the numerical technique employed.

> The extracted polytropic exponent γ appears approximately constant across the dataset and is close to 1. This suggests that the process is nearly isothermal, consistent with the assumption of constant temperature during the experiment.

> The variation in γ across interpolation methods is $\Delta\gamma \approx 0.00131$.

SUMMERY

In this assignment, I used numerical interpolation and extrapolation techniques to reconstruct physical information from experimental data of a piston–cylinder system. Since the pressure sensor produced a nonlinear voltage output instead of direct pressure readings, I first calibrated the sensor by constructing a numerical relation between voltage and pressure.

After obtaining the calibration curve, I used it to estimate pressures both within and outside the calibration range. By comparing extrapolated values with reference data, I analyzed how errors increase beyond the measured region and understood the limitations of extrapolation.

Next, I converted voltage measurements into pressure during piston compression and reconstructed the pressure–volume (P–V) relation of the gas. Finally, I investigated whether the gas follows a polytropic relation and numerically estimated the effective exponent γ.

Overall, this assignment demonstrated how numerical methods are essential in interpreting experimental data and how the choice of technique can influence the physical conclusions drawn from the data.

I really enjoyed doing this Assignment:)